

Companion to Bird & de Moor, Algebra of Programming

§5. Datatypes in Allegories

§5.1. Relators

(5.1a)

Every hom-set of an allegory is a poset, so an allegory is a **locally posetal 2-category**: the 2-cell from R to S IS $R \sqsubseteq S$. A **relator** $F : \mathcal{C} \rightarrow \mathcal{D}$ is a 2-functor between allegories:

$$F(1) = 1 \quad F(RS) = F(R)F(S) \quad R \sqsubseteq S \implies F(R) \sqsubseteq F(S)$$

Preserving $\circ$ is **not** asked for — $\circ$ is an identity-on-objects involution $\mathcal{C}^{\circ} \rightarrow \mathcal{C}$, no part of the 2-category.

(5.1b)

the statement

- 1 $\text{Map}(f) \implies \text{Map}(F(f))$ and $\text{Map}(f) \implies F(f^\circ) = F(f)^\circ$. Lemma 5.1
- 2 Over a **tabular** allegory a functor is a relator $\iff$ it preserves $\circ$. Theorem 5.1 the book's reverse direction (preserving $\circ \implies$ relator) needs the source to be **Rel**: the counterexample over a tabular allegory that is not **Rel** shows it can fail
- 3 $F(R^\circ) = F(R)^\circ$ for every R , so $F(R)^\circ$ needs no bracket.
- 4 Two relators agreeing on maps are equal. Corollary 5.1
- 5 $\text{Coreflexive}(X) \wedge \text{Coreflexive}(Y) \implies F(X \cap Y) = F(X) \cap F(Y)$ Ex 5.2
- 6 $F(R \cap S) \sqsubseteq F(R) \cap F(S)$, and strictly. Ex 5.2
- 7 $F(\text{dom}(R)) = \text{dom}(F(R))$ Ex 5.5

The **power relator** $P \multimap x s P(R) y s \iff (\forall a \in x s. \exists b \in y s. a R b) \wedge (\forall b \in y s. \exists a \in x s. a R b)$ — is where the fourth is strict: for $R = \{(a_1, b_1), (a_2, b_2)\}$ and $S = \{(a_1, b_2), (a_2, b_1)\}$ the pair $(\{a_1, a_2\}, \{b_1, b_2\})$ is in $P(R) \cap P(S)$, while $R \cap S = \emptyset$.

§5.2. Relational products

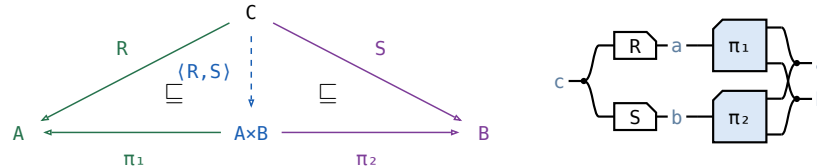
(5.1)

The **fork** of $R : \mathcal{C} \rightarrow \mathcal{A}$ and $S : \mathcal{C} \rightarrow \mathcal{B}$ is $\langle R, S \rangle \triangleq R \pi_1^\circ \cap S \pi_2^\circ$, where (π_1, π_2) is the tabulation of τ .

(5.2a)

$$\langle R, S \rangle \pi_1 = \text{dom}(S) R \quad (5.6) \quad \langle R, S \rangle \pi_2 = \text{dom}(R) S \quad (5.7)$$

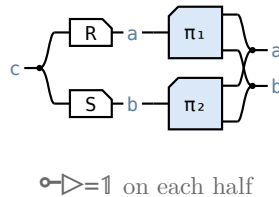
(5.2b)



A domain is coreflexive, so $\langle R, S \rangle \pi_1 \sqsubseteq R$, with equality exactly when S is entire; for maps both triangles commute and $\langle f, g \rangle$ is unique. In **Rel**, $c \in \langle R, S \rangle (a, b)$ iff $c R a$ and $c S b$ — copy c , then R on one strand and S on the other, which is $\triangleleft (R \otimes S)$ on the right.

No $\circ$ survives the translation. $\pi_1 = 1 \otimes \bullet$ discards the second component, so $\pi_1^\circ = 1 \otimes \bullet$ **creates** one out of nothing, and $\cap$ is copy, run both, merge. Draw that and the created strands — the two dots with no left end, and the crossing they force — are merged against real ones, which is the monoid's unit law:

(5.2c)

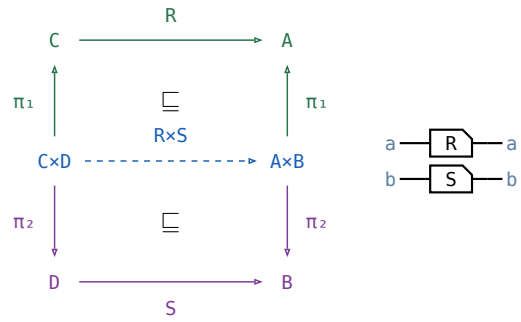


§5.2.1. Relational product $R \times S$

(5.2)

$R \times S \triangleq \langle \pi_1 R, \pi_2 S \rangle$, a relator in each argument but no longer a categorical product.

(5.2.1a)



Right-then-up is $(R \times S)\pi_1$, up-then-right is $\pi_1 R$, and $(R \times S)\pi_1 \sqsubseteq \pi_1 R$, equality when S is entire. In $\mathbf{Rel}$, $(c, d) \in (R \times S) \iff (a, b) \in R$ and $d \in S \iff b$ — two strands side by side, no copy dot: $R \times S = R \otimes S$.

§5.2.2. Absorption

For $X : E \rightarrow C$ and $Y : E \rightarrow D$, $\langle X, Y \rangle (R \times S) = \langle XR, YS \rangle$. Both sides are this picture:



(5.3)

(5.2.2a)

	the law	picture
1	$\langle X, Y \rangle (R, S)^\circ = (X R^\circ) \cap (Y S^\circ)$	
2	$\langle R, S \rangle^\circ \langle U, V \rangle \sqsubseteq (R^\circ U) \times (S^\circ V)$	
3	$\text{Map}(f) \implies f \langle R, S \rangle = \langle f R, f S \rangle$ f a map; it fails for an arbitrary arrow;	
4	$X \langle R, S \rangle \sqsubseteq \langle X R, X S \rangle$ the half of the row above that holds for every X	
5	$F(R \times S) \text{ unzip}(F) \sqsubseteq \text{unzip}(F) (F(R) \times F(S))$ $\text{unzip}(F) \triangleq \langle F(\pi_1), F(\pi_2) \rangle$; only $\sqsubseteq$ for an arbitrary R, S	
6	$g = \text{curry}(f) \iff (g \times 1) \text{ eval} = f$ reading $\times$ as the relational product, does $\mathbf{Rel}$ have exponentials?	

§5.3. Relational coproducts

(5.3a)

	the statement	picture
	$R : a_1 \rightarrow c, S : a_2 \rightarrow c$	
1	$[R, S] \triangleq (l^\circ R) \cup (r^\circ S)$ (5.9) The tape is the union — a particle entering at $A+B$ takes exactly one branch — and the two mirrored boxes are what makes the branches disjoint.	
2	$[R, S] = [\frac{R}{\exists}, \frac{S}{\exists}] \exists$	
3	$l [R, S] = R, r [R, S] = S$ $l T = R \wedge r T = S \implies T = [R, S]$	
4	$U : a_1 \rightarrow d, V : a_2 \rightarrow d$ $[U, V]^\circ [R, S] = (U^\circ R) \cup (V^\circ S)$ (5.11)	
5	$Z : c \rightarrow d$ $[R, S] Z = [R Z, S Z]$	

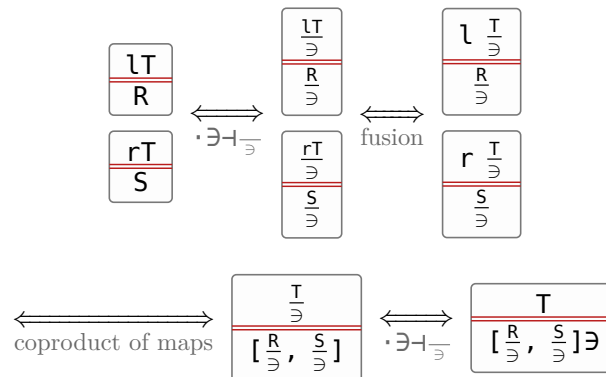
(5.3b)

	the statement	picture
	$R : a_1 \rightarrow b_1, S : a_2 \rightarrow b_2$	
1	$R + S \triangleq [R l, S r]$ (5.10)	
2	$l l^\circ = 1, r r^\circ = 1$ Ex 5.12	$a_1 \text{---} l \text{---} a \text{---} l \text{---} a_1 = a_1 \text{---} a_1$ $a_2 \text{---} r \text{---} a \text{---} r \text{---} a_2 = a_2 \text{---} a_2$
3	$l r^\circ = 0, r l^\circ = 0$ Ex 5.12	$a_1 \text{---} l \text{---} a \text{---} r \text{---} a_2 = a_1 \text{---} 0 \text{---} a_2$
4	$(l^\circ l) \cup (r^\circ r) = 1$ Ex 5.12	

§5.3.1. $[R, S] = [\frac{R}{\exists}, \frac{S}{\exists}] \exists$ — a case of relations is the case of their transposes, then $\exists$

The universal-property row is not free: l, r were only ever asked to be a coproduct of **maps**. They stay one once every arrow is allowed because $\exists$ sends an arrow $A \rightarrow C$ to a map $A \rightarrow PC$ reversibly, so the map coproduct can be applied underneath it. For any $T : A+B \rightarrow C$,

(5.3.1a)



(5.3.1b)



Nothing here holds only up to $\cong$: every triangle commutes on the nose, which is the difference from the fork above. The border spells $[R, S] = [\frac{R}{S}, \frac{S}{S}] \bar{\exists}$, and pushing $\bar{\exists}$ into the union that $[\cdot, \cdot]$ on maps already is turns that back into the definition, $(\bar{l} \circ \frac{R}{S} \cup \bar{r} \circ \frac{S}{S}) \bar{\exists} = \bar{l} \circ R \cup \bar{r} \circ S$.

(5.3.1c)

the law	picture
1 prove (5.11), and say why duality does not carry it over from the product law	
2 $(R + S) \cap ([P, Q] [U, V]^\circ)$ $= (R \cap (P \cup^\circ)) + (S \cap (Q \cup^\circ))$ $[P, Q] [U, V]^\circ$ is a full 2×2 of composites; $R+S$ is diagonal, so the meet cuts the two off-diagonal branches	The diagram shows the composition of relations R, S, P, Q, U, and V. It illustrates the relationship between the meet of R and S and the union of P and Q, and how this relates to the composition of U and V. The diagram is divided into two main parts by an equals sign, showing that the composition of the meet and union is equal to the meet of the compositions.

§5.4. The power relator $P(R)$

(5.4a)

For $R : A \rightarrow B$,

$$P(R) \triangleq (\epsilon \setminus (R \epsilon)) \cap ((\exists R) / \exists)$$

$$E(R) \triangleq \frac{\exists R}{\exists} = ((\exists R) / \exists) \cap (\exists / (\exists R))^\circ$$

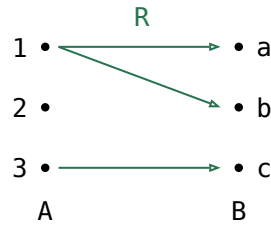
$$xs \ P(R) \ ys \iff (\forall a \in xs. \exists b \in ys. a \ R \ b) \wedge (\forall b \in ys. \exists a \in xs. a \ R \ b)$$

Every element of xs is related by R to some element of ys , and conversely.

(5.4b)

	in Rel	in words	$\text{im}(xs)$	partner
1	$(\exists R) / \exists$	$\forall y \in ys. \exists x \in xs. x \ R \ y$	$ys \subseteq \text{im}(xs)$	every y has a partner
2	$(\exists / (\exists R))^\circ$	$\forall y. (\exists x \in xs. x \ R \ y) \rightarrow y \in ys$	$ys \supseteq \text{im}(xs)$	ys leaves out no partner
3	$E(R)$	$ys = \{y \mid \exists x \in xs. x \ R \ y\}$	$ys = \text{im}(xs)$	every y has a partner, and none is left out
4	$((\exists R^\circ) / \exists)^\circ$	$\forall x \in xs. \exists y \in ys. x \ R \ y$	—	every x has a partner
5	$P(R)$	$P(R)(xs, ys) \iff (\forall y. y \in ys \implies (\exists x. x \in xs \wedge R(x, y))) \wedge (\forall x. x \in xs \implies (\exists y. y \in ys \wedge R(x, y)))$	—	every x and every y has a partner

(5.4c)



	xs	$(\exists R) / \exists$	$(\exists / (\exists R))^\circ$	$E(R)$	$((\exists R^\circ) / \exists)^\circ$	$P(R)$
1	$\emptyset$	$\emptyset$	all 8 subsets of abc	$\emptyset$	all 8 subsets of abc	$\emptyset$
2	$\{1\}$	$\emptyset, a, b, ab$	ab, abc	ab	a, b, ab, ac, bc, abc	a, b, ab
3	$\{2\}$	$\emptyset$	all 8 subsets of abc	$\emptyset$	none	none
4	$\{3\}$	$\emptyset, c$	c, ac, bc, abc	c	c, ac, bc, abc	c
5	$\{1, 2\}$	$\emptyset, a, b, ab$	ab, abc	ab	none	none
6	$\{1, 3\}$	all 8 subsets of abc	abc	abc	ac, bc, abc	ac, bc, abc
7	$\{2, 3\}$	$\emptyset, c$	c, ac, bc, abc	c	none	none
8	$\{1, 2, 3\}$	all 8 subsets of abc	abc	abc	none	none

(5.4d)

	law	the reading
1	$X \sqsubseteq P(R) \iff X \exists \sqsubseteq \exists R \wedge X^\circ \exists \sqsubseteq \exists R^\circ$	One containment, and the same one at R° — which is the definition read off the two divisions. Hence $P(R^\circ) = P(R)^\circ$, and $R \sqsubseteq S \implies P(R) \sqsubseteq P(S)$.
2	$P(1) = 1$	The straightness axiom verbatim: extensionality is P 's unit law.
3	$\text{Map}(f) \implies P(f) = E(f)$	In Rel, $xs \ P(f) \ ys \iff ys = \{f(a) \mid a \in xs\}$. The half at f° says every $a \in xs$ has its $f(a)$ on ys ; f has just the one image per a , so that already says ys contains everything xs reaches, which is the fraction's second half. For a map the two definitions coincide.
4	$P(R \ S) = P(R) \ P(S)$	$\sqsupseteq$ is the division cancellation laws. $\sqsubseteq$ is the one law in this section that is not a calculation: it needs a tabulation of $P(RS)$.

§5.4.1. Type relator

(5.4.1a)

Let F be a binary relator with initial type (α, T) , so T is a type functor. $F(R, S)$ is its action on a pair, and $F(X)$ abbreviates $F(1, X)$, the F of the reduce section. For every object A the initial algebra is $\alpha : F(A, TA) \rightarrow TA$, among the maps. T acts on an arrow $R : A \rightarrow B$ by

$$T(R) = \langle F(R, 1) \ \alpha \rangle$$

(5.4.1b)

		name	law	what it says
1		the defining equation	$T(R) = \langle F(R, 1) \ \alpha \rangle$	Rebuild the structure with α , applying R to the parameter on the way.
2		type relator	$T(R)^\circ = T(R^\circ)$	A datatype acts on relations, not only on maps — the map of the converse is the converse of the map.

§5.5. Relational catamorphisms

(5.5a)

let F be a relator and has **initial algebra** $\alpha : F(T) \rightarrow T$ in the subcategory of functions. α is also initial in the allegory:

§5.5.1. The defining equation

(5.5.1a)



An F -algebra is a **weakened** natural transformation. A transformation $F \Rightarrow 1$ on $\mathcal{A}$ would need a component $FX \rightarrow X$ at every object and a commuting square at every arrow, but F -Algebra only need it works on T and A .

(5.5.1b)

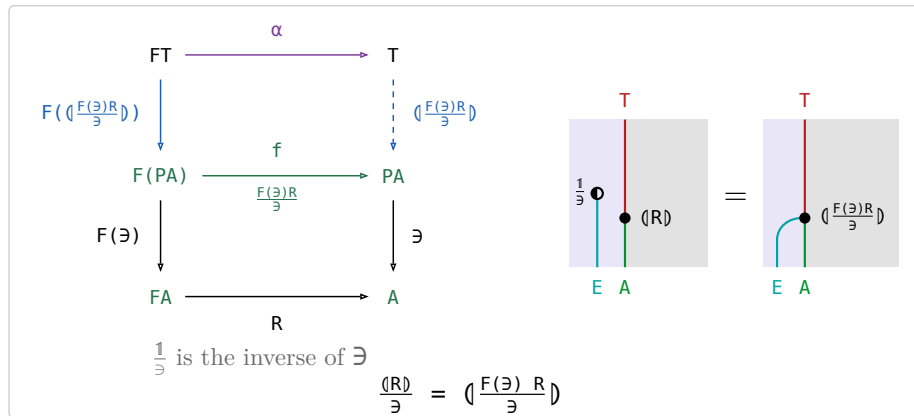
$$\langle [c, f] \rangle = \text{reduce}(c, f)$$

catamorphism: arrows of the form $\langle f \rangle$ are called catamorphisms, and these two examples reveal the notion to be a familiar idea in abstract clothing

1	part	Nat	[A]
2	datatype	$\text{Nat} ::= \text{zero} \mid \text{succ Nat}$	$[A] ::= \text{nil} \mid \text{cons}(A, [A])$
3	base functor F	$F(X) = 1 + X$	$F(X) = 1 + A \times X$
4	initial algebra α	$\alpha = [\text{zero}, \text{succ}]$ $: 1 + \text{Nat} \rightarrow \text{Nat}$	$\alpha = [\text{nil}, \text{cons}]$ $: 1 + A \times [A] \rightarrow [A]$
5	the fold, pointwise	$\langle [c, f] \rangle(\text{zero}) = c$ $\langle [c, f] \rangle(\text{succ}(n)) = f(\langle [c, f] \rangle(n))$	$\langle [c, f] \rangle(\text{nil}) = c$ $\langle [c, f] \rangle(\text{cons}(a, x)) = f(a, \langle [c, f] \rangle(x))$

§5.5.2. $\langle R \rangle = \langle \frac{F(\exists) R}{\exists} \rangle \exists$ — folding with R is a fold of maps into sets, then $\exists$

(5.5.2a)



(5.12)

$$\begin{array}{c}
 \boxed{\frac{\alpha X}{F(X)R}} \xleftrightarrow{\cdot \exists \vdash \frac{1}{\exists}} \boxed{\frac{\frac{\alpha X}{\exists}}{\frac{F(X)R}{\exists}}} \xleftrightarrow{\cdot \exists \vdash \frac{1}{\exists}} \boxed{\frac{\frac{\alpha X}{\exists}}{\frac{F(\frac{X}{\exists})R}{\exists}}} \\
 \xleftrightarrow{\text{relator, fusion twice}} \boxed{\frac{\frac{\alpha}{\frac{X}{\exists}}}{\frac{F(\frac{X}{\exists})}{\frac{F(\exists)R}{\exists}}}} \xleftrightarrow{\text{reduce of maps}} \boxed{\frac{\frac{X}{\exists}}{\langle \frac{F(\exists)R}{\exists} \rangle}} \xleftrightarrow{\cdot \exists \vdash \frac{1}{\exists}} \boxed{\frac{X}{\langle \frac{F(\exists)R}{\exists} \rangle \exists}}
 \end{array}$$

the last two rows at $X := \langle R \rangle$:

$\alpha X = F(X) R \iff \frac{X}{\exists} = \langle \frac{F(\exists) R}{\exists} \rangle$ (5.5.2b)					
in a tabular allegory, with F a relator and $\alpha : F(T) \rightarrow T$ its initial algebra: a relation X out of T satisfies the fold equation $\alpha X = F(X) R$ of R exactly when $\frac{X}{\exists}$ is the fold of the map $\frac{F(\exists) R}{\exists}$					
$\frac{X}{\exists}$	$\frac{\alpha^\circ F(X) R}{\exists}$	$\alpha^\circ \frac{F(X) R}{\exists}$	$\alpha^\circ \frac{F(\frac{X}{\exists}) R}{\exists}$	$\alpha^\circ F(\frac{X}{\exists}) \frac{F(\exists) R}{\exists}$	$\langle \frac{F(\exists) R}{\exists} \rangle$
(1)	(2)	(3)	(4)	(5)	(6)
1	$\frac{X}{\exists}$	α iso; $\alpha X = F(X) R$			
2	$= \frac{\alpha^\circ F(X) R}{\exists}$	α° is a map because α is an iso, so it leaves Λ : $\frac{\alpha^\circ F(X) R}{\exists} = \alpha^\circ \frac{F(X) R}{\exists}$ — $\text{Map}(f)$			
3	$= \alpha^\circ \frac{F(X) R}{\exists}$	$\implies \frac{f R}{\exists} = f \frac{R}{\exists}$			
4	$= \alpha^\circ \frac{F(\frac{X}{\exists}) R}{\exists}$	$X = \frac{X}{\exists} \exists \implies \frac{R}{\exists} \exists = R$			
5	$= \alpha^\circ F(\frac{X}{\exists}) \frac{F(\exists) R}{\exists}$	$F(\frac{X}{\exists}) = F(\frac{X}{\exists}) F(\exists)$ as F is a functor, and $F(\frac{X}{\exists})$ is a map (a relator sends maps to maps), so it leaves Λ — $\triangleq$ relator, $\text{Map}(f)$			
6	$= \langle \frac{F(\exists) R}{\exists} \rangle$	$\implies \frac{f R}{\exists} = f \frac{R}{\exists}$			
		fold uniqueness, $\frac{F(\exists) R}{\exists}$ being a map — $(h : f \rightarrow g) \triangleq f h = F(h) g$, $\text{Map}(f)$			
		$\implies \alpha \langle f \rangle = F(\langle f \rangle) f$			

§5.6. Combinatorial functions

(5.6a)

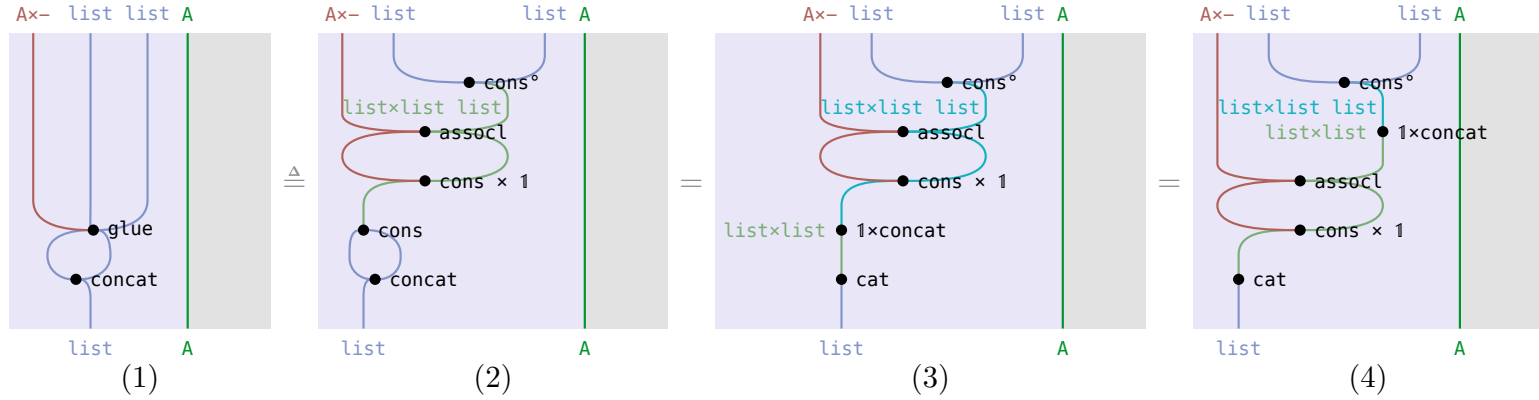
	definition	type	note
1	$[A] ::= \text{nil} \mid \text{cons}(A, [A]) \quad \text{list}(A) = [A]$	$\mathcal{A} \rightarrow \mathcal{A}$	The list type, under the short name it keeps.
2	$\text{list}(R) = ([\text{nil}, (R \times 1) \text{ cons}])$	$[A] \rightarrow [B]$	The relator's action on $R : A \rightarrow B$: one R per element, the shape untouched.
3	$\text{subseq} = ([\text{nil}, \text{cons} \cup \pi_2])$	$[A] \rightarrow [A]$	$xs \text{ subseq } ys$: ys is xs with elements dropped — cons keeps the head, π_2 drops it.
4	$\text{prefix} = ([\text{nil}, \text{cons} \cup \multimap \text{ nil}])$ $\text{prefix} = \text{cat}^\circ \pi_1 = \text{init}^*$	$[A] \rightarrow [A]$	ys is an initial segment of xs ; the first nil is where it stops early. $\text{init} \triangleq \text{snoc}^\circ \pi_1$.
5	$\text{suffix} = \text{cat}^\circ \pi_2 = \text{tail}^*$	$[A] \rightarrow [A]$	The dual, $\text{tail} \triangleq \text{cons}^\circ \pi_2$; as a reduce it needs snoc -lists.
6	$\text{partition} = \text{concat}^\circ$	$[A] \rightarrow [[A]]$	This cat is restricted to $[A]^* \times [A] \rightarrow [A]$, so ys is a list of non-empty segments of xs .
7	$\text{concat} = ([\text{nil}, \text{cat}])$	$[[A]] \rightarrow [A]$	Joins the segments back up, which is why its converse splits a list.
8	inits	$[A] \rightarrow [[A]]$	Implements $\frac{\text{prefix}}{\exists}$, listing the prefixes by increasing length.
9	tails	$[A] \rightarrow [[A]]$	Implements $\frac{\text{suffix}}{\exists}$ by decreasing length — the opposite order.
10	$\text{cpp}, \text{cpl}, \text{cpr}$	$[A] \times [B] \rightarrow [A \times B]$ $[A] \times B \rightarrow [A \times B]$ $A \times [B] \rightarrow [A \times B]$	Every pair (a, b) with a from the first list and b from the second; cpl , cpr keep one side a single element.
11	$\text{cpp setify} = (\text{setify} \times \text{setify}) \frac{\exists \times \exists}{\exists}$ $\text{cpl setify} = (\text{setify} \times 1) \frac{\exists \times 1}{\exists}$ $\text{cpr setify} = (1 \times \text{setify}) \frac{1 \times \exists}{\exists}$	$[A] \times [B] \rightarrow P(A \times B)$	Forgetting the order with setify , cpp is the transpose of $\exists \times \exists$.
12	$\text{cp}(F) \triangleq \frac{F(\exists)}{\exists}$	$F(PA) \rightarrow PFA$	$\text{cp}(F)$, the transpose of a relator's action on $\exists$; cpp implements it at $F(A) = A \times A$.
13	$\text{cp}(\text{list}) = ([\frac{[\text{nil}, (\exists \times \exists) \text{ cons}]}{\exists}])$ $\frac{[\text{nil}, (\exists \times \exists) \text{ cons}]}{\exists} = [\text{nil} \frac{1}{\exists}, \frac{\exists \times \exists}{\exists}]$ $E(\text{cons})]$	$\text{list}(PA) \rightarrow P(\text{list}(A))$	$\text{cp}(\text{list})$: (5.12) $(R) = ([\frac{F(\exists)}{\exists} R]) \exists$ at $\text{list}(\exists) = ([\text{nil}, (\exists \times 1) \text{ cons}])$, then the algebra expanded.
14	$\text{cplist} = ([[\text{nil} \text{ wrap}, \text{cpp list}(\text{cons})]])$	$[[A]] \rightarrow [[A]]$	$\text{cp}(\text{list})$ with every set a list: cpp in place of $\frac{\exists \times \exists}{\exists}$.

glue concat $\sqsubseteq (1 \times \text{concat}) \text{ cons}$

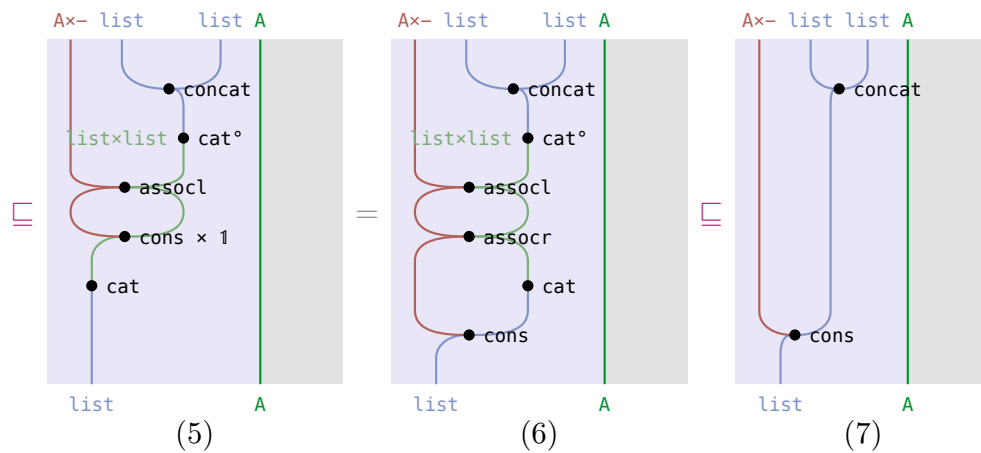
(5.6b)

putting **a** on the front of the first segment and flattening (**glue concat**) gives at most flattening the segments and putting **a** on the front of the result (**(1×concat) cons**)

the starting composite



1	glue concat	definition of glue
2	$\triangleq (1 \times \text{cons}^\circ) \text{ assocl } \text{cons} \times 1 \text{ cons } \text{concat}$	since $\text{concat} = ([\text{nil}, \text{cat}])$
3	$= (1 \times \text{cons}^\circ) \text{ assocl } \text{cons} \times 1 (1 \times \text{concat}) \text{ cat}$	naturality of assocl
4	$= (1 \times (\text{cons}^\circ (1 \times \text{concat}))) \text{ assocl } \text{cons} \times 1 \text{ cat}$	since $\text{concat} = ([\text{nil}, \text{cat}])$



5	$\sqsubseteq (1 \times \text{concat}) (1 \times \text{cat}^\circ) \text{ assocl } \text{cons} \times 1 \text{ cat}$	since $(\text{cons} \times 1) \text{ cat} = \text{assocr } (1 \times \text{cat}) \text{ cons}$
6	$= (1 \times \text{concat}) (1 \times \text{cat}^\circ) \text{ assocl } \text{assocr } (1 \times \text{cat}) \text{ cons}$	since $\text{assocl } \text{assocr} = 1$ and $\text{cat}^\circ \text{ cat} \sqsubseteq 1$
7	$\sqsubseteq (1 \times \text{concat}) \text{ cons}$	

§5.6.1. $\frac{\text{subseq}}{\exists} = ([\text{nil } \frac{1}{\exists}, (\frac{1 \times \exists}{\exists} \text{ E}(\text{cons}), \pi_2) \text{ cup}])$ — the set of subsequences is a fold of maps into sets

(5.6.1a)

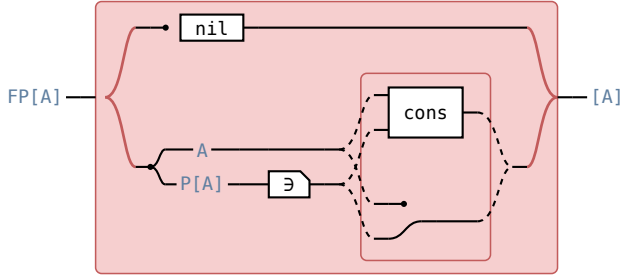
$\text{cup} \triangleq \frac{\pi_1 \exists \cup \pi_2 \exists}{\exists} : \text{PA} \times \text{PA} \rightarrow \text{PA}$, so $\frac{R \cup S}{\exists} = (\frac{R}{\exists}, \frac{S}{\exists}) \text{ cup}$.

$$\frac{F(\exists) [\text{nil}, \text{cons } u \pi_2]}{\exists} = [\text{nil } \frac{1}{\exists}, \frac{(1 \times \exists) (\text{cons } u \pi_2)}{\exists}] \quad (5.6.1b)$$

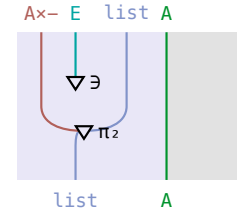
the set of lists the algebra builds is, from nothing, just **nil**, and from a head and a set of tails, every tail in the set with the head put on or left off — (5.12) $\langle R \rangle = \langle \frac{F(\exists)}{\exists} R \rangle \exists$ at **subseq** = $\langle [\text{nil}, \text{cons } u \pi_2] \rangle$, **subseq** = $\langle [\text{nil}, \text{cons } u \pi_2] \rangle$.

circuit — the fork is $F([A])=1+A \times [A]: \text{nil}$ above, the pair below

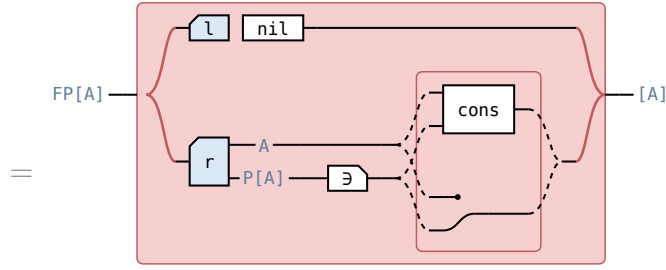
Hinze–Marsden



$$\frac{F(\exists) [\text{nil}, \text{cons } u \pi_2]}{\exists}$$

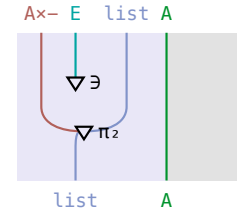


the π_2 operand of $\text{cons } u \pi_2$ under the $1 \times \exists$ summand of $F(\exists)$, i.e. $(1 \times \exists) \pi_2$

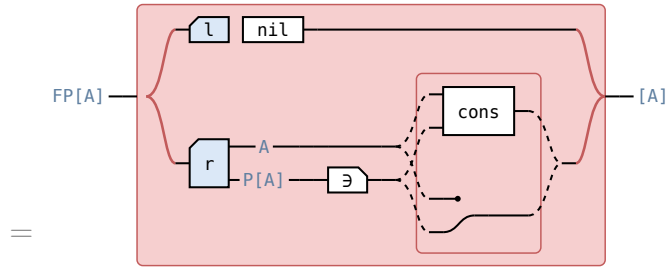


$$\frac{(1+1 \times \exists) [\text{nil}, \text{cons } u \pi_2]}{\exists}$$

$$F(X)=1+A \times X \text{ — list}(A) = [A]$$

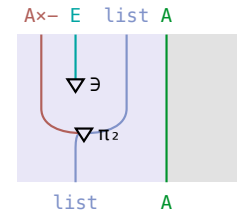


the same operand under $1+1 \times \exists$, whose $1 \times \exists$ summand it sits in

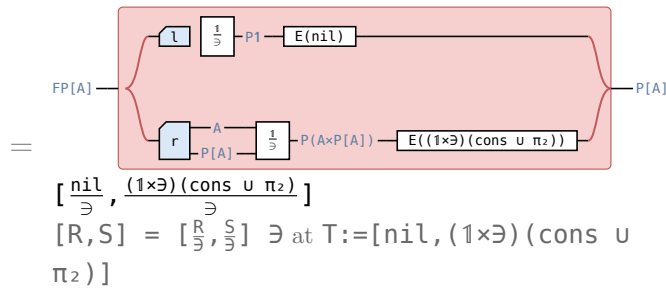


$$\frac{[\text{nil}, (1 \times \exists) (\text{cons } u \pi_2)]}{\exists}$$

$$R+S \triangleq [Rl, Sr], l[R, S]=R, r[R, S]=S \text{ — } [R, S] \triangleq (l \circ R) \cup (r \circ S), l[R, S]=R, r[R, S]=S, l T=R \wedge r T=S \Rightarrow T=[R, S]$$

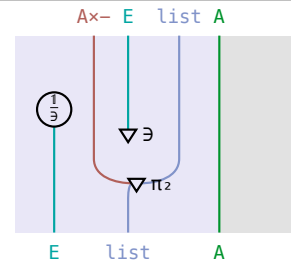


the π_2 operand of the second arm $(1 \times \exists) (\text{cons } u \pi_2)$

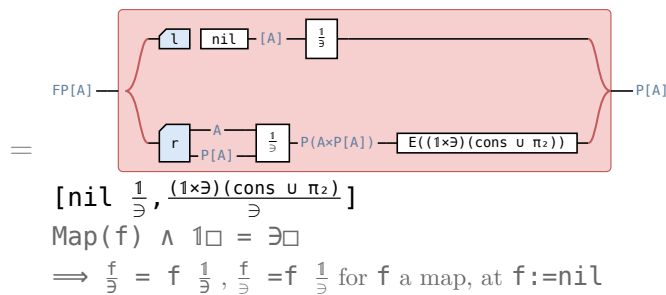


$$\frac{[\text{nil}, (1 \times \exists) (\text{cons } u \pi_2)]}{\exists}$$

$$[R, S] = [\frac{R}{\exists}, \frac{S}{\exists}] \exists \text{ at } T := [\text{nil}, (1 \times \exists) (\text{cons } u \pi_2)]$$



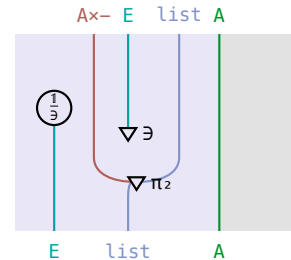
the π_2 operand under its $\frac{1}{\exists}$, the arm $(1 \times \exists) \pi_2 = \pi_2 \exists$ rewrites



$$\frac{[\text{nil } \frac{1}{\exists}, (1 \times \exists) (\text{cons } u \pi_2)]}{\exists}$$

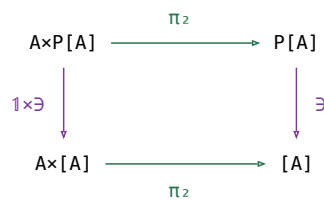
$$\text{Map}(f) \wedge 1 \square = \exists \square$$

$$\Rightarrow \frac{f}{\exists} = f \frac{1}{\exists}, \frac{f}{\exists} = f \frac{1}{\exists} \text{ for } f \text{ a map, at } f := \text{nil}$$



the same operand; the two rows differ only in the **nil** arm

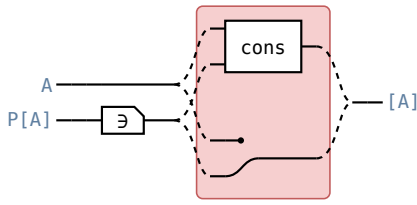
(5.6.1c)



$$\frac{(1 \times \exists) (\text{cons} \cup \pi_2)}{\exists} = \left(\frac{1 \times \exists}{\exists} E(\text{cons}), \pi_2 \right) \text{cup} \quad (5.6.1d)$$

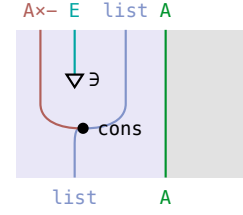
power transpose of join: the power transpose of the join of two relations is $\left(\frac{R}{\exists}, \frac{S}{\exists} \right) \text{cup}$, where **cup** is the function that returns the union of two sets

circuit

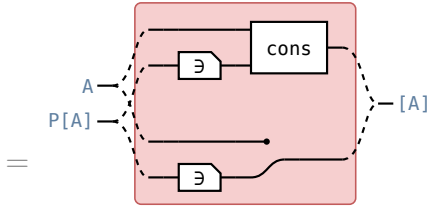


$$\frac{(1 \times \exists) (\text{cons} \cup \pi_2)}{\exists} = \frac{F(\exists) [\text{nil}, \text{cons} \cup \pi_2]}{\exists} = [\text{nil} \frac{1}{\exists}, \frac{(1 \times \exists) (\text{cons} \cup \pi_2)}{\exists}] \text{'s second branch}$$

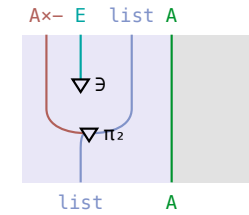
Hinze–Marsden



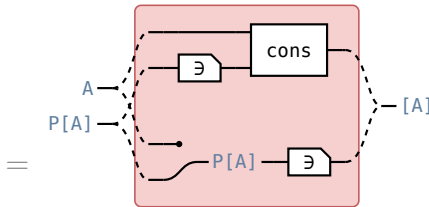
the **cons** operand of **cons** $\cup$ π_2



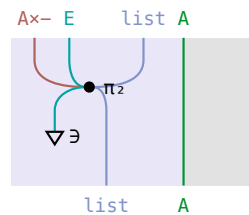
$$\frac{(1 \times \exists) \text{cons} \cup (1 \times \exists) \pi_2}{\exists} = T(X_1 \cup X_2) = TX_1 \cup TX_2 \text{ — } (S \cup T) R = (S R) \cup (T R), R(S \cup T) = (R S) \cup (R T), R \cap (S \cup T) = R \cap S \cup R \cap T, (R \cup S)^\circ = S^\circ \cup R^\circ$$



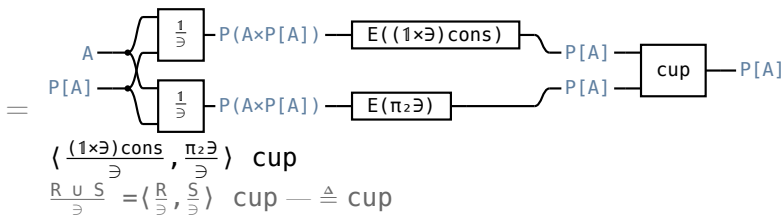
the π_2 operand of $(1 \times \exists) \text{cons} \cup (1 \times \exists) \pi_2$



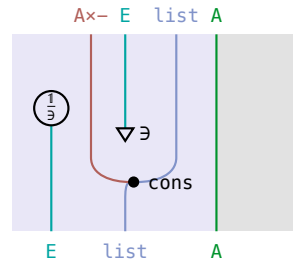
$$\frac{(1 \times \exists) \text{cons} \cup \pi_2 \exists}{\exists} = (1 \times \exists) \pi_2 = \pi_2 \exists \text{ — } (R \times S) \pi_1 \sqsubseteq \pi_1 R \wedge (R \times S) \pi_2 \sqsubseteq \pi_2 S \text{ at } \pi_2, \text{ an equality because } 1 \text{ is entire}$$



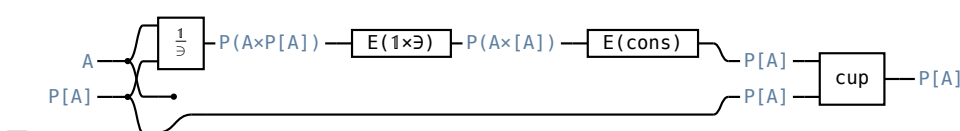
the π_2 operand of $(1 \times \exists) \text{cons} \cup \pi_2 \exists$



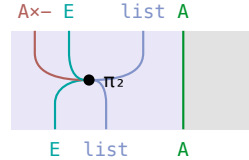
$$\left(\frac{(1 \times \exists) \text{cons}}{\exists}, \frac{\pi_2 \exists}{\exists} \right) \text{cup} = \frac{R \cup S}{\exists} = \left(\frac{R}{\exists}, \frac{S}{\exists} \right) \text{cup} \text{ — } \triangleq \text{cup}$$



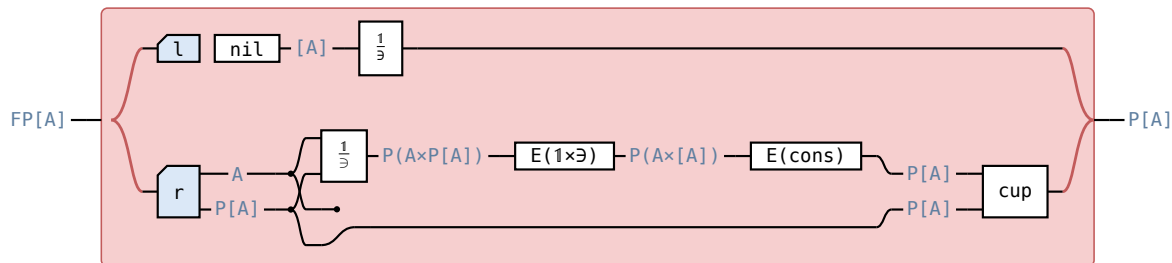
the **cons** operand under its $\frac{1}{\exists}$



$$\left(\frac{1 \times \exists}{\exists} E(\text{cons}), \pi_2 \right) \text{cup} = \frac{S}{\exists} E(R) = \frac{S}{\exists} R, \text{ absorption } \frac{S}{\exists} E(R) = \frac{SR}{\exists} \text{ at } S := 1 \times \exists, R := \text{cons}; \text{ fusion and } \frac{\exists}{\exists} = 1 \text{ on the } \pi_2 \text{ operand}$$



the π_2 operand, bare π_2



$$\left[\frac{\text{nil}}{\exists}, \left(\frac{1 \times \exists}{\exists} E(\text{cons}), \pi_2 \right) \text{cup} \right]$$

which writes $P\text{cons}$; **cons** is a map, and there $P(\text{cons}) = E(\text{cons}) \text{ — Map}(f) \Rightarrow P(f) = E(f)$.

(5.6.1e)

§5.7. Lax natural transformations (LaT)

(5.7a)

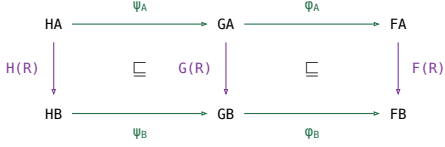
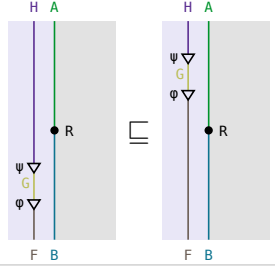
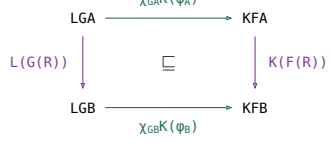
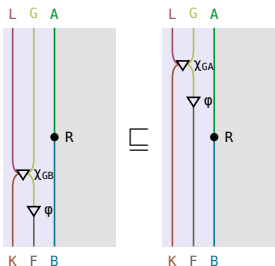
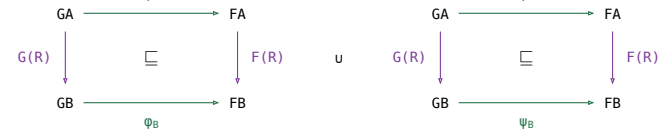
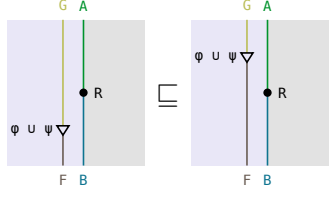
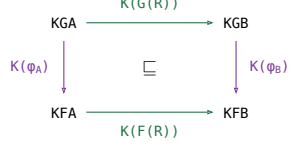
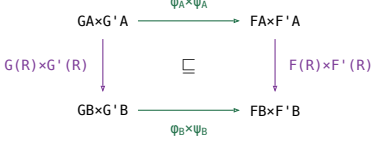
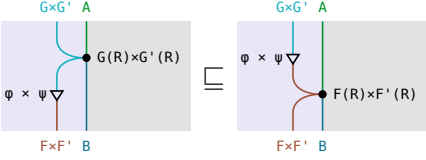
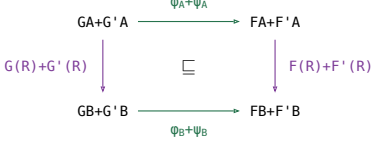
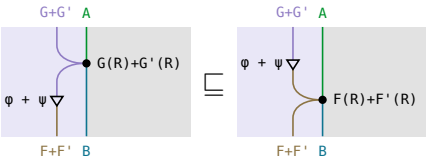
For relators $G, F : \mathcal{C} \rightarrow \mathcal{D}$ and components $\varphi_A : GA \rightarrow FA$, φ is **lax at** $R : A \rightarrow B$ when $G(R)\varphi_B \sqsubseteq \varphi_A F(R)$ — both sides $GA \rightarrow FB$, the left through the component at B , the right through the one at A .

$\varphi : G \Rightarrow F$ is a **lax natural transformation** (LaT) when it is lax at **every** R . (5.13)

Lax at every **map** already gives LaT, and at a map the inequation is an equality $G(f)\varphi = \varphi F(f)$: laxness is about relations only.

(5.7b)

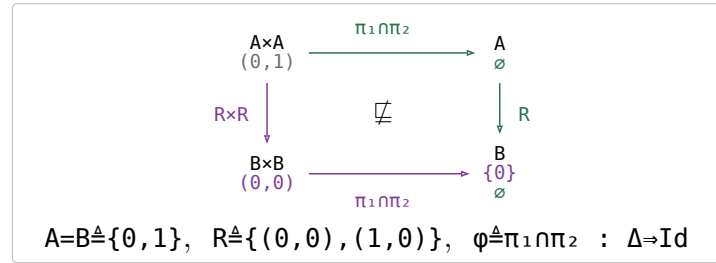


	closed under	commutative diagram	string diagram
1 composition $\psi\phi$		 $H(R)\psi_B \sqsubseteq \psi_A G(R)$ and $G(R)\phi_B \sqsubseteq \phi_A F(R)$ give $H(R)(\psi_B \phi_B) \sqsubseteq (\psi_A \phi_A) F(R)$	
2 horizontal composition $\chi \circ \phi$		 $\phi : G \Rightarrow F$ with $G, F : \mathcal{C} \rightarrow \mathcal{D}$ and $\chi : L \Rightarrow K$ with $L, K : \mathcal{D} \rightarrow \mathcal{E}$ give $\chi \circ \phi : L \circ G \Rightarrow K \circ F$, the family $A \mapsto \chi_{GA} K(\phi_A)$ $L(G(R)) \chi_{GB} \sqsubseteq \chi_{GA} K(G(R))$ is χ lax at $G(R)$; then $K(G(R)\phi_B) \sqsubseteq K(\phi_A F(R))$ is K applied to ϕ's own inequation; $\chi := \mathbb{1}_K$ gives $K(\phi) : K \circ G \Rightarrow K \circ F$, K composed on the outside; $\phi := \mathbb{1}_G$ gives $\chi G : L \circ G \Rightarrow K \circ G$, G composed on the inside two candidates, $\chi_{GA} K(\phi_A)$ and $L(\phi_A) \chi_{FA}$, ordered by $\sqsubseteq$; this row is the first	
3 union $\phi \cup \psi$		 $G(R)\phi_B \sqsubseteq \phi_A F(R)$ and $G(R)\psi_B \sqsubseteq \psi_A F(R)$ give $G(R)(\phi_B \cup \psi_B) \sqsubseteq (\phi_A \cup \psi_A) F(R)$	
4 a relator K $K(\phi)$		 $G(R)\phi_B \sqsubseteq \phi_A F(R)$ gives $K(G(R))K(\phi_B) \sqsubseteq K(\phi_A)K(F(R))$	
5 product $\phi \times \psi$		 $\phi : G \Rightarrow F$ and $\psi : G' \Rightarrow F'$ give $\phi \times \psi : G \times G' \Rightarrow F \times F'$ $(R \times S)(U \times V) = (RU) \times (SV)$ and monotonicity in both slots — the row above's K applied to the inequation, run on a bifunctor; the fork is the derived case $\langle \phi, \psi \rangle = \triangleleft (\phi \times \psi)$, and its ONE cost is the lax copy law $R \triangleleft \sqsubseteq \triangleleft (R \times R) \rightarrow \triangleleft (R \otimes R), R \multimap \sqsubseteq \multimap$	
6 coproduct $\phi + \psi$		 $\phi : G \Rightarrow F$ and $\psi : G' \Rightarrow F'$ give $\phi + \psi : G + G' \Rightarrow F + F'$ $(R + S)(U + V) = (RU) + (SV)$ and monotonicity in both slots; the co-fork is the derived case $[\phi, \psi] = (\phi + \psi) \nabla$, and ∇ costs nothing the step it would need is $(\phi_A F(R)) \cap (\psi_A F'(R)) \sqsubseteq (\phi \cap \psi)_A F(R)$, the wrong direction of R $(S \cap T) \sqsubseteq (R S) \cap (R T)$ a counterexample: $aa = (\emptyset, 1) \wedge bb = (\emptyset, 0) \wedge (R \times R)(aa, bb) \wedge b_1 = \{\emptyset\} \wedge b_1 = E(\pi_1 \cap \pi_2)\{bb\} \wedge a_1 = \emptyset \wedge a_1 = E(\pi_1 \cap \pi_2)\{aa\} \wedge b_2 = \emptyset \wedge b_2 = E(R)(a_1)$ $\Rightarrow \neg((R \times R)(\pi_1 \cap \pi_2) \sqsubseteq (\pi_1 \cap \pi_2) R)$	
7 meet — fails $\phi \cap \psi$			
8 Each row is the general slide $R_A X \sqsubseteq X' R_B$ and $R_B Y \sqsubseteq Y' R_C$ give $R_A (XY) \sqsubseteq (X' Y') R_C$ at $R_A, R_B := G(R), F(R)$ and $X, X' := \phi_B, \phi_A$, quantified over R ; at $X' = X$ with R_A, R_B endorelations it is the monotonicity reading instead — the $\forall R$ sits outside the law, and is the only difference			

§5.7.1. meet is not closed in LaT

counterexample — $\varphi, \psi : \mathbf{G} \Rightarrow \mathbf{F}$ lax natural does NOT give $\varphi \cap \psi$ lax natural

(5.7.1a)



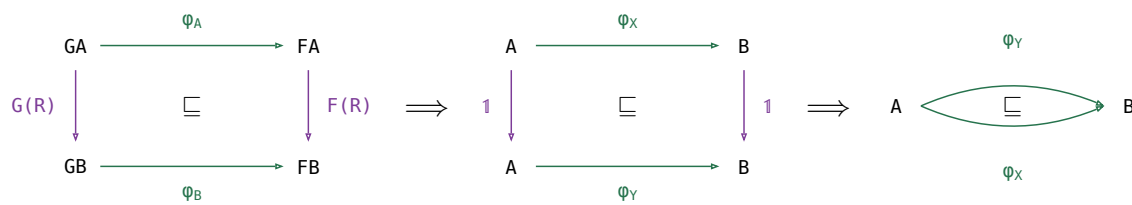
§5.7.2. Two stacked towers

(5.7.2a)

	lower — inside ONE allegory	upper — between allegories
1 0-cell	an object A	an allegory $\mathcal{C}$ — a whole lower tower
2 1-cell	a relation $R : A \rightarrow B$	a relator $F : \mathcal{C} \rightarrow \mathcal{D}$
3 2-cell	$R \sqsubseteq S$	a LaT $\varphi : \mathbf{G} \Rightarrow \mathbf{F} \triangleq \text{lax}$
4 order	$R \sqsubseteq S$	$\varphi \sqsubseteq \psi$ componentwise
5 union	$R \cup S$	$\varphi \cup \psi$, survives $G(R) (\varphi_B \cup \psi_B) \sqsubseteq (\varphi_A \cup \psi_A) F(R)$
6 product	$R \times S \triangleq \times$	$\varphi \times \psi$, survives; a TENSOR, not a categorical product; SYMMETRIC $\langle R, S \rangle$ $\pi_1 = \text{dom}(S) \ R, \langle R, S \rangle \ \pi_2 = \text{dom}(R) \ S \ (G(R) \times G'(R)) (\varphi_B \times \psi_B) \sqsubseteq (\varphi_A \times \psi_A) (F(R) \times F'(R))$
7 coproduct	$R + S$	$\varphi + \psi$, survives; a BIPRODUCT $(G(R) + G'(R)) (\varphi_B + \psi_B) \sqsubseteq (\varphi_A + \psi_A) (F(R) + F'(R))$
8 meet	$R \cap S$	$\varphi \cap \psi$ componentwise, fails $aa = (0,1) \wedge bb = (0,0) \wedge (R \times R)(aa, bb) \wedge b_1 = \{0\} \wedge b_1 = E(\pi_1 \cap \pi_2)\{bb\} \wedge a_1 = \emptyset \wedge a_1 = E(\pi_1 \cap \pi_2)\{aa\} \wedge b_2 = \emptyset \wedge b_2 = E(R)(a_1) \Rightarrow \neg((R \times R) (\pi_1 \cap \pi_2) \sqsubseteq (\pi_1 \cap \pi_2) R)$
9 converse	R°	φ° , fails, oplax: $\varphi_A^\circ G(R) \sqsubseteq F(R) \varphi_B^\circ$
10 zero object	z with $1 \ z = \emptyset$	the constant relator at such a z , initial and terminal

§5.7.3. RES is a special case of LaT

(5.7.3a)

 $G \triangleq \text{const } A$, $F \triangleq \text{const } B$ — the relators $X \rightarrow A$ and $X \rightarrow B$, each sending every arrow to $1 \triangleq \text{lax}$ with $\mathcal{C}$ two objects and one non-identity arrow $X \rightarrow Y$, a LaT $\text{const } A \Rightarrow \text{const } B$ is the pair $\varphi_X, \varphi_Y : A \rightarrow B$ with $\varphi_Y \sqsubseteq \varphi_X$ — the datum RES not every LaT is constant: $\exists : \mathbf{P} \Rightarrow \text{Id} \ X \sqsubseteq P(R) \iff X \exists \sqsubseteq \exists R \wedge X^\circ \exists \sqsubseteq \exists R^\circ \ \pi_2 : (A \times -) \Rightarrow \text{Id} \ (1 \times \exists) \ \pi_2 = \pi_2 \exists$ $\triangleleft : \text{Id} \Rightarrow \Delta \ R \triangleleft \leq \triangleleft (R \otimes R), R \multimap \leq \multimap$